

Supernova bounds on quadratically coupled scalars: $\phi\phi$ pair emission with energy-momentum conservation, the trace channel, pions and transport

Technical note v2.1, Gravity Innovation project, 27 September 2026. Unrefereed draft for an expert check. Supersedes v1 (26 September 2026); v2.1 replaces the rough ΔN_{eff} estimate of Section 8 with a Boltzmann calculation.

Provenance. This analysis was carried out with AI assistance (Claude). Two independent derivations (A and B; scripts in `supernova/sn_A/`, `supernova/sn_B/` of the repository) were reconciled in v1. For v2 an internal referee report was addressed and the key inputs were re-derived with realistic potentials: the trace-channel matrix elements were evaluated exactly at finite ω with the AV18, Reid93 and Nijmegen-II potentials (`supernova/sn_deep/pw_solver.py`, `trace_tables.py`, `trace_thermal.py`, `trace_degeneracy.py`), cross-checked against a leading-order chiral potential (`chiral_lo_check.py`), and the transport was done by Monte Carlo (`transport_mc.py`, `diffusion_1d.py`). Every number in this note comes from those scripts or from the v1 scripts, re-run for v2; where v2 changes a v1 number, the change is stated. The note has not been refereed. We are asking a supernova or nuclear theorist to check Sections 3, 5 and 6, in that order. Code and outputs: <https://github.com/paulwilkerson1985-cmd/gravity-innovation>. Contact: Paul Wilkerson (please open an issue on this repository).

Abstract

We compute the energy loss of a proto-neutron star into pairs of a light scalar ϕ with a quadratic matter coupling, and state which operator is bounded. For the **universal (conformal) coupling** $\mathcal{L} = -(\phi^2/2M^2)T^\mu{}_\mu$ (the symmetron; also a $d_V = 2$ scalar unparticle), energy-momentum conservation gives the exact identity $\Theta(q) = (k_i k_j / \omega^2 - \delta_{ij})T^{ij}(q)$: $NN \rightarrow NN\phi\phi$ measures only the stress spectral function of the medium, and the $O(m_N/\omega)$ radiation assumed in the 2008 estimate of Olive and Pospelov (OP) is absent. The rate splits into a traceless (quadrupole) channel fixed by on-shell NN data, the pair analogue of the Kaluza-Klein-dilaton result of Hanhart et al. and of the 2007 unparticle estimates, and a trace channel fixed at $\omega \rightarrow 0$ by the Wigner time delay, $X_{ii} = (1 + k \partial_k)T_{NN}$. We verify this theorem with the exact finite- ω matrix element $\langle f|2V + rV'|i \rangle$ for AV18, Reid93 and Nijmegen-II (to 0.5% at 10 MeV and 0.1% above 75 MeV), and find that the pair phase space samples $\omega \sim E$, where the exact element is **1.8-1.9** $\times$ the soft form: the trace/traceless ratio is $K_T^{\text{MB}} = 12$ (10-13), reduced by Pauli blocking to **6.7-8.1** at the Raffelt point. Pion absorption $\pi^- p \rightarrow n\phi\phi$ has no conservation suppression and dominates for $Y_{\pi^-} \gtrsim 0.2\%$; its systematics are the largest uncertainty. Monte Carlo transport replaces the semi-trapping guess of v1.

For the universal coupling, $L_\phi(1s) < 3 \times 10^{52}$ erg/s gives $M \gtrsim 6$ TeV (5.3 TeV on the cold profile, 7.9 on the hot; 3.5-6.0 without pions; floor 2.2-3.9 from the traceless channel alone), and the excluded region is a band from 1-2 TeV up to M_{min} . For **OP's literal operator** $-(\phi^2/2M^2)m_N \bar{N}N$, which couples neither to pions nor to the nuclear force, the trace channel is $\langle f|V|i \rangle$ ($K_T^{\text{MB}} = 2.8$ -5.3) and the pion channel is **0.22** $\times$, giving $M \gtrsim 4.5$ TeV (3.8-5.8), $\approx 0.72 \times$ the universal value. Applied to either operator, the 2008 estimate overstates the emissivity by 10-100 $\times$. A Boltzmann calculation of the thermal relic (Section 8) gives $\Delta N_{\text{eff}} = 0.24$ (0.14-0.32) at $M = 5$ TeV and 0.17 (0.10-0.27) at 6 TeV; the 2026 BBN+CMB+BAO limit $\Delta N_{\text{eff}} < 0.107$ then excludes $M \lesssim 7.5$ TeV (5.9-9.6) for the universal coupling if the universe reheated above a few GeV. That bound is stronger but depends on the thermal history; the supernova bound does not. The bound holds for the symmetron sign of the coupling; for the opposite sign (Balkin et al.) dense matter sources ϕ and the induced linear coupling gives $M \gtrsim 10^7$ TeV. The conservation-law suppression was noted in 2007 (Section 9); we have not found an earlier calculation of $\phi\phi$ pair emission that combines it with the trace channel, pions and transport.

1. Which operator is bounded

We consider a scalar coupled to matter through the Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$, $A = 1 + \phi^2/2M^2$ (Hinterbichler-Khoury symmetron). To first order,

$$\mathcal{L}_{\text{univ}} = -\frac{\phi^2}{2M^2}\Theta, \quad \Theta \equiv T^\mu{}_\mu \rightarrow m_N \bar{N}N + m_\pi^2 \pi^2 + \dots \text{ (on shell)}, \quad (1.1)$$

and a uniform A rescales every hadronic scale ($m_N, m_\pi, f_\pi, \Lambda_{\text{QCD}}$), so the two-nucleon potential transforms as $V_A(r) = AV(Ar)$. OP instead couple a dimensionless φ to $B_j(\varphi)m_j \bar{\psi}_j \psi_j$ for $j = n, p, e$ and to F^2 , with $B_j = 1 + \xi_j \varphi^2/2$, $\xi_n \simeq \xi_p = 1$ "mostly induced by the gluon B -function" (their eqs. 2.2-2.4). With $\phi = M_* \varphi$,

$$\mathcal{L}_{\text{OP}} = -\frac{\phi^2}{2M_*^2} m_N \bar{N}N (+e, F^2 \text{ terms}). \quad (1.2)$$

The two agree for free nucleons ($m_N \bar{N}N = \Theta$ on shell up to $O(p^2/m_N)$) but differ in the medium: (1.2) does not couple to pions or to the interaction energy of the nuclear force.

	universal (1.1)	OP (1.2)
traceless (quadrupole) channel	identical	identical
trace-channel matrix element	$\langle f 2V + rV' i \rangle$ (time-delay theorem at $\omega \rightarrow 0$)	$\langle f V i \rangle$
$\pi^- p \rightarrow n\phi\phi$	pion-line diagram dominates	nucleon legs only (0.22 $\times$)
$\gamma\gamma \rightarrow \phi\phi$	trace anomaly, $\xi_F \sim \alpha/\pi$	ξ_F free

v1 compared its bound on (1.1) with OP's bound on (1.2); here both are given. Conventions: OP's M_* ($\xi_N = 1$) = M ; Stadnik-Flambaum's $\Lambda' = \sqrt{2}M$; Balkin et al.'s $M_\phi = M$ with the *opposite sign*. For the symmetron sign dense matter restores the symmetric phase ($\langle \phi \rangle = 0$, $m_{\text{eff}} \approx 36 \text{ keV (TeV/M)}(\rho/3 \times 10^{14} \text{ g cm}^{-3})^{1/2} \ll T$) and only pair processes occur; for the opposite sign dense matter sources ϕ and the induced linear coupling gives $M_\phi \gtrsim 10^7$ TeV (Appendix B).

2. Pair emission and the conservation identity

From $\partial_\mu T^{\mu\nu} = 0$, for pair four-momentum $q = (\omega, \mathbf{k})$, $\kappa = |\mathbf{k}|/\omega \leq 1$,

$$\Theta(q) = \left(\frac{k_i k_j}{\omega^2} - \delta_{ij} \right) T^{ij}(q) \equiv P_{ij} T^{ij}(q). \quad (2.1)$$

The identity is exact for the full conserved $T^{\mu\nu}$ between exact many-body eigenstates; evaluating it between two-nucleon scattering states is the quasiparticle approximation, whose in-medium corrections (m^* , T -matrix, LPM) are the residual NN uncertainty. The monopole (baryon number) and dipole (momentum; cancels for equal n, p couplings) are absent; B's relativistic one-pion-exchange check confirms the cancellation diagram by diagram ($|\mathcal{M}_{\text{rel}}|^2/(2m)^4|\Theta_{\text{NR}}|^2 = 1.00 \pm 0.01$ at $p = 30\text{--}60$ MeV). Treating the massless pair as one scalar of invariant mass $s = \omega^2(1 - \kappa^2)$ with measure $d\mathbf{s}/(32\pi^2 M^4)$,

$$Q = \frac{1}{64\pi^4 M^4} \int d\omega \omega^4 \int_0^1 \kappa^2 d\kappa \langle S_\Theta(\omega, \kappa) \rangle, \quad \langle |P:X|^2 \rangle_k = \kappa^4 \frac{2}{15} X^T : X^T + (1 - \frac{\kappa^2}{3})^2 |X_{ii}|^2, \quad (2.2)$$

with no cross term; the κ integrals give the weights $2/105$ (traceless) and $68/315$ (trace). The traceless channel, $X^T = T_{NN}(\mathbf{k}'\mathbf{k}' - \mathbf{k}\mathbf{k})^T/\mu\omega$, is fixed by $\sigma^{(2)} = \int d\Omega (d\sigma/d\Omega) \sin^2 \theta$; it is HPRS's KK-dilaton theorem (their eq. 38) summed over a continuum of masses ($\int d\mathbf{s} \kappa^5 \propto 1/7$) and the quadrupole term of the 2007 unparticle papers at $d_U = 2$ (Section 9). Because the ω^4 phase space pushes emission to $\chi = \omega M/\bar{p}^2 \sim 1$, the $O(\chi^0)$ terms that HPRS note are "not constrained by on-shell data and conservation" dominate: that is the trace channel. *Unparticle map*: a $d_U = 2$ scalar operator has the flat spectral density $A_2 d^4 P/(2\pi)^4$, $A_2 = 1/8\pi$, i.e. a free massless pair; matching to $(\lambda_S/\Lambda_U) \bar{N} N O_U$ gives $\lambda_S/\Lambda_U = m_N/\sqrt{2}M^2$.

3. The trace channel: time-delay theorem and finite- ω evaluation

This is the one new theorem the rest hangs on, and it can be checked in an afternoon. Between exact NN eigenstates with $E_f = E_i - \omega$ the source is the scale derivative of the two-body Hamiltonian, $S = dH/dA|_{A=1} = 2V + \mathbf{r} \cdot \nabla V$ (kinetic term $\rightarrow A^{-2}$, potential $\rightarrow AV(A\mathbf{r})$), so

$$\langle f|X_{ii}|i \rangle \propto \langle f|2V + \mathbf{r} \cdot \nabla V|i \rangle, \quad (3.1)$$

exact to all orders in V . In the soft limit observables depend on k/A only, and Hellmann-Feynman gives

$$X_{ii}|_{\omega \rightarrow 0} = (1 + k \partial_k) T_{NN}|_\theta, \quad (1 + k d/dk) f_\ell = e^{2i\delta_\ell} \delta'_\ell(k), \quad \sigma_{\text{tr}} \equiv \int d\Omega |X_{ii}|^2 = w\pi \sum_J (2J+1) \frac{1}{4} \text{Tr} |dS_J/dk|^2 \quad (3.2)$$

($w = 1$ for np , 2 for nn): the Wigner time delay. It vanishes in the unitary, scale-invariant limit, as it must, and is the dilatation-operator term of the soft-dilaton theorem (Di Vecchia et al.) applied to nonrelativistic scattering. It needs only the on-shell S -matrix but assumes $V_A(\mathbf{r}) = AV(A\mathbf{r})$, i.e. universal rescaling including Λ_{QCD} : true for (1.1), not for (1.2).

Check with realistic potentials. We solved the coupled-channel Schrödinger equation for AV18, Reid93 and Nijmegen-II (all partial waves to $J = 4$; PWA93 np phases reproduced to $\lesssim 0.5^\circ$ below 300 MeV) and evaluated (3.1) between eigenchannel standing waves on an $E_f \times E_i$ grid. On the diagonal the exact element reproduces (3.2) to 0.5% at $E_{\text{cm}} = 10$ MeV and 0.1% above 75 MeV, identically for the three potentials and for nn, np (Table C1); the residual is the finite-difference step. A leading-order chiral potential with a different short-range form gives the same identity (1.000–1.005). The trace/elastic ratio is $c_{\text{tr}}^2 = 3.0\text{--}3.1$ (nn), 2.7–2.8 (np) for $T_{\text{lab}} = 150\text{--}350$ MeV (v1: 2.3–3.1 from approximate tables).

Finite ω . The pair phase space samples $\omega \sim E$, where the exact element exceeds the soft form: per matrix element $1.1\text{--}1.2\times$ at $\omega/E = 0.5$ and $1.7\text{--}2.1\times$ at $\omega/E = 0.8$ (AV18). Thermally (Maxwell-Boltzmann, $\rho = 3 \times 10^{14} \text{ g cm}^{-3}$, $Y_p = 0.3$), with $K_T \equiv Q_{NN}/Q_{\text{traceless}}$:

T [MeV]	K_T v1 ($B, \omega \rightarrow 0$)	K_T ($\omega \rightarrow 0$ at $\bar{k}$, potentials)	K_T exact finite ω (AV18 / Reid93 / NijmII)	finite/soft
20	6.4	6.7	11.6 / 11.8 / 11.7	1.86–1.90
30	6.4	6.8–7.2	12.2 / 12.3 / 12.1	1.82–1.92
40	5.9	6.7–7.6	12.1 / 11.7 / 11.6	1.69–1.86

The spread across potentials is $\leq 3\%$, and holding σ_{tr} fixed above $E_{\text{cm}} = 175$ MeV changes it by $\leq 3\%$. This confirms A's potential-model range 8–14 and supersedes v1's $K_T = 6.5$ (4–14), which was B's soft-limit value. A local LO chiral potential ($R_0 = 1.0$, 1.2 fm; contacts fitted at 10 MeV) gives a thermal S -wave finite/soft ratio of 2.1–2.5 against 3.0–3.1 for AV18; S waves carry 85–90% of the finite- ω trace channel, so LO-EFT-like short-range dynamics would give $K_T^{\text{MB}} \approx 10$. **Adopted: $K_T^{\text{MB}} = 12$ (10–13).** An $N^2\text{LO}/N^3\text{LO}$ evaluation with data-quality phases would settle the last $\sim 20\%$.

Degeneracy. The enhancement lives at large ω (slow final nucleons) and is Pauli-blocked more strongly than the soft form: Fermi-Dirac/Maxwell-Boltzmann ratios at (30 MeV, 3×10^{14} , $Y_p = 0.3$) are 0.91–1.00 (traceless; B's v1 value 0.85 was Monte Carlo noise), 0.74–0.89 (soft trace) and 0.55–0.76 (finite- ω trace). The effective ratio is $K_T^{\text{FD}} = 6.3$, 8.1, 9.1 at $T = 20, 30, 40$ MeV, 7.0 and 8.4 at the cold- and hot-profile temperature peaks (Table C2). With the LO-EFT-like lower edge, $K_T^{\text{FD}} = 6.7\text{--}8.1$ at the Raffelt point and 7–10 in the emitting zones. The residual trace-channel uncertainty is now the in-medium physics, not the ω dependence.

OP's operator. Replacing $2V + \mathbf{r} \cdot \nabla V$ by V in the same code (AV18) gives $K_T^{\text{MB}} = 5.3/3.6/2.8$ at $T = 20/30/40$ MeV (finite/soft = 0.75/0.42/0.27); for (1.2) the finite- ω corrections *reduce* the trace channel.

4. NN emissivity

At $\rho = 3 \times 10^{14} \text{ g cm}^{-3}$, $Y_p = 0.3$, $M = 1$ TeV, with the AV18 finite- ω trace channel and Fermi-Dirac statistics (erg $\text{g}^{-1}\text{s}^{-1}$; inputs in Appendix C):

T [MeV]	ϵ_{OP}	ϵ_T^{FD} (traceless)	$\epsilon_{\text{tr}}^{\text{FD}}$ (trace)	K_T^{FD}	$\epsilon_{NN}^{\text{FD}}$	v1 (6.5 ϵ_T)	M_{min} (Raffelt): OP / traceless / NN [TeV]
20	1.15×10^{23}	7.9×10^{19}	4.1×10^{20}	6.3	4.9×10^{20}	5.1×10^{20}	10.4 / 1.7 / 2.65
30	4.76×10^{23}	6.6×10^{20}	4.7×10^{21}	8.1	5.4×10^{21}	4.3×10^{21}	14.8 / 2.9 / 4.8
40	1.30×10^{24}	3.0×10^{21}	2.4×10^{22}	9.1	2.7×10^{22}	1.9×10^{22}	19.0 / 4.2 / 7.2

A fit to nine grid points ($T = 20\text{--}42$ MeV, $\rho = (1\text{--}4.4) \times 10^{14}$, $Y_p = 0.12\text{--}0.3$; nn_emissivity_new.json), good to $\sim 15\%$:

$$\epsilon_{NN}^{\text{FD}} \simeq 5.4 \times 10^{21} \text{ erg g}^{-1}\text{s}^{-1} \left(\frac{\rho}{3 \times 10^{14}} \right)^{0.75} \left(\frac{T}{30 \text{ MeV}} \right)^{5.7} g(Y_p) \left(\frac{\text{TeV}}{M} \right)^4, \quad g = 0.40 \left(\frac{1 - Y_p}{0.7} \right)^2 + 0.51 \frac{Y_p(1 - Y_p)}{0.21} + 0.09 \left(\frac{Y_p}{0.3} \right)^2. \quad (4.1)$$

At identical conditions OP's estimate overstates the NN emissivity by $\approx 700\times$ (traceless only) or $\approx 90\times$ (total, 30 MeV). The NN -only Raffelt-point bound is **4.6–4.8 TeV** (v1: 4.6).

5. Pion-induced emission and its systematics

A thermal π^- that is absorbed gives up its rest energy and its scalar charge ($\langle \pi | \Theta | \pi \rangle \ni 2m_\pi^2$), so $\pi^- p \rightarrow n \phi \phi$ has no conservation suppression and the pair carries $\langle \omega \rangle \approx 170$ –210 MeV. A and B evaluate the same tree-level Lagrangian in two field bases and agree to $\leq 5\%$ before final-state blocking (Appendix A); in A's basis the pion-line diagram dominates, and for OP's operator only the nucleon legs remain, $0.22\times$. With B's computed final-neutron blocking factor $B(T)$,

$$\epsilon_\pi \simeq 5.9 \times 10^{22} B(T) \text{ erg g}^{-1} \text{ s}^{-1} \left(\frac{Y_\pi}{1\%} \right) \left(\frac{Y_p}{0.3} \right) \left(\frac{\rho}{3 \times 10^{14}} \right) \left(\frac{T}{30 \text{ MeV}} \right)^{1.9} \left(\frac{\text{TeV}}{M} \right)^4, \quad B = 0.39, 0.59, 0.70 \text{ at } T = 20, 30, 40 \text{ MeV}. \quad (5.1)$$

v1 used a flat $B = 0.7$. At (30 MeV, 3×10^{14}), $Q_\pi/Q_{NN} \approx 6.4 (Y_\pi/1\%)$, so pions dominate for $Y_\pi \gtrsim 0.2\%$. The transport runs of Section 6 used the flat factor; (5.1) lowers the cold-profile central bound by $\approx 3\%$ (that core emits at $T \approx 25$ –31 MeV) and leaves the hot one unchanged, while the two-body channel below raises both by $\approx 3\%$, so the Section 6 values stand to $\pm 3\%$.

Systematics, ranked. (1) *Abundance*. Ideal gas: $Y_\pi = 0.1$ –0.5%; virial (Fore-Reddy 2020): 1–5%; Carenza et al. 2021: 1.1% at 1 s. Fore, Kaiser, Reddy and Warrington (2024) find in chiral perturbation theory that the π^- mass rises with density in neutron-rich matter, the opposite of the p -wave softening behind the larger abundances; we keep Y_π –(37 MeV) $\in [0.3, 3]\%$, central 1%, shape $\exp[-(m_\pi - 80 \text{ MeV})(1/T - 1/37)]$, but the range is skewed low. $M_{\min} \propto Y_\pi^{1/4}$: $\times 0.75$ –1.3. Y_π also peaks later than 1 s. (2) *Blocking*, 0.39–0.70 in (5.1). (3) *In-medium propagator*. The dominant pion-line diagram propagates an off-shell pion with $|q| \sim 150$ MeV, where the Δ -hole self-energy is $\mathcal{O}(1)$; the open item on the rate per pion, $\times 0.5$ –2. (4) *Two-body absorption bremsstrahlung*. $\pi^- n p \leftrightarrow n n$ and $NN \leftrightarrow NN\pi$ change the scalar charge by $\Delta S \approx 290 \text{ MeV} - \omega$ and are not conservation-suppressed; a sudden-approximation estimate with pionic-atom widths ($\Gamma_{\text{abs}} \approx 50 \text{ MeV}$ at 3×10^{14}) gives $5.6 \times 10^{21} (Y_\pi/1\%) \text{ erg g}^{-1} \text{ s}^{-1}$ at 1 TeV, **13% of the one-body channel** ($+3\%$ in M); not in the tables. (5) *Others*. $\pi N \rightarrow \pi N \phi \phi \sim 7\%$; $\pi\pi \rightarrow \phi\phi$ (vertex $2m_\pi^2 + s$, $n_\pi/n_p \approx 4 \times 10^{-4}$): a few per cent, not v1's 10^{-3} ; $\Delta(1232)$: $\lesssim 1\%$ (no tree-level Δ pole, Θ being diagonal in the baryon basis); $e^\pm$: $(m_e/T)^2$; $\gamma\gamma$: $\xi_F \sim \alpha/\pi$.

6. Transport

The only ϕN vertex in the symmetric phase is the contact term, $\sigma_{\phi N} = m_N^2/4\pi M^4 = 2.7 \times 10^{-41} \text{ cm}^2 (\text{TeV}/M)^4$, giving a central optical depth $\tau_c \approx 30$ at $M = 4 \text{ TeV}$ and ≈ 2 at 8 TeV. v1 applied a hand factor $M_{\min} \times (0.85\text{--}0.95)$. v2 uses a Monte Carlo (transport_mc.py): pairs produced from $Q(r) = Q_{NN} + Q_\pi$ with the channel spectra ($\langle \omega_{\text{pair}} \rangle = 6T$ for NN , 190 MeV for pions) are random-walked through the profile with elastic contact scattering (full kinematics against Maxwellian nucleons, so the spectrum degrades toward the local T) and detailed-balance re-absorption, $\Gamma_{\text{abs}} = Q/u_{BB}$. A flux-limited diffusion solve without degradation (diffusion_1d.py) covers $M = 1$ –10 TeV.

profile, channels	M [TeV]	τ_c	$f_{\text{esc}} = L_{\text{esc}}/L_{\text{prod}}$	$\langle E_{\text{esc}} \rangle / \langle E_{\text{prod}} \rangle$	$M_{\min}, L_{\text{esc}} < 3 \times 10^{52}$ (free-streaming)	$< 5.7 \times 10^{52}$
cold, NN only	4 / 5 / 6	33 / 13 / 6.5	0.64 / 0.73 / 0.84	0.67 / 0.74 / 0.84	3.5 (4.1)	none for $M \geq 2.5$ (free-streaming 3.5)
cold, $NN + \pi$ (1%)	4 / 5 / 6	33 / 13 / 6.5	0.50 / 0.68 / 0.77	0.63 / 0.71 / 0.78	5.3 (5.8)	4.3 (5.0)
hot, NN only	4 / 5 / 6	29 / 12 / 5.7	0.60 / 0.71 / 0.82	0.63 / 0.72 / 0.82	6.0 (6.3)	4.9 (5.4)
hot, $NN + \pi$ (1%)	5 / 6 / 8	12 / 5.7 / 1.8	0.75 / 0.84 / 0.94	0.78 / 0.84 / 0.94	7.9 (8.1)	6.7 (6.9)

Energy degradation dominates; re-absorption (0.2 of the pairs at 4 TeV on the cold profile with pions, < 0.01 above 6 TeV) matters only below $M \approx 4 \text{ TeV}$. The self-consistent bounds are $\times 0.92$ (cold) to 0.98 (hot) of the free-streaming values for the central case and $\times 0.85$ –0.95 for NN only, confirming and sharpening the v1 guess.

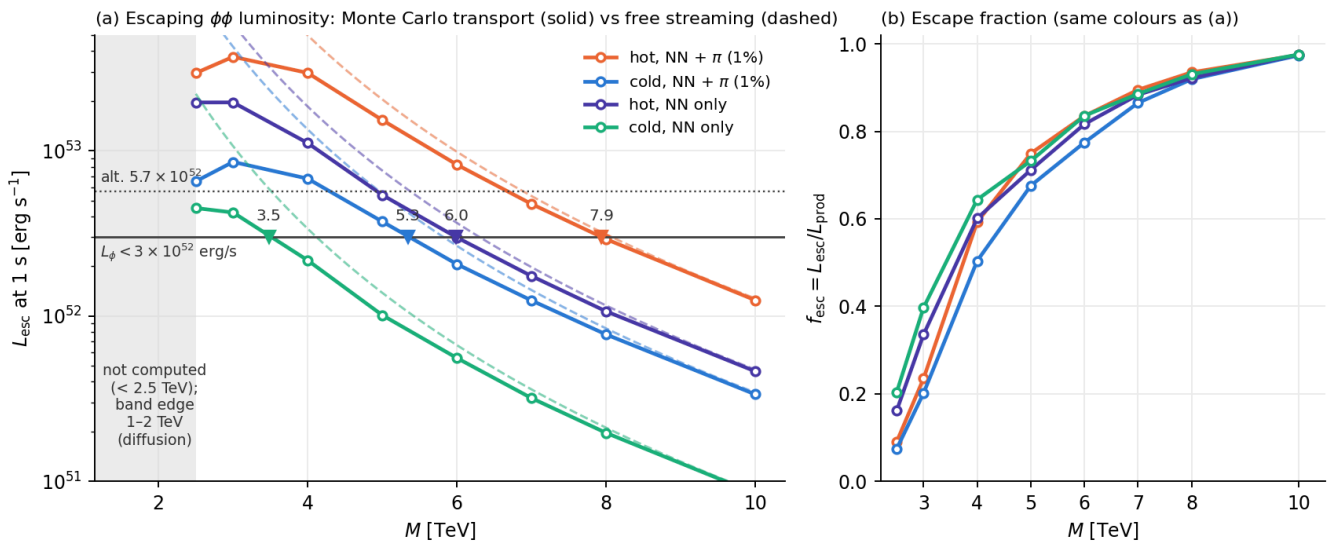


Figure 1. (a) Escaping $\phi\phi$ luminosity at 1 s from the Monte Carlo transport (solid, points) and for free streaming L_{prod}/M^4 (dashed), for the cold and hot profiles with and without pions ($Y_\pi = 1\%$), against $L_\phi < 3 \times 10^{52} \text{ erg/s}$ (solid line; 5.7×10^{52} dotted). Triangles mark the self-consistent $M_{\min}$; the shaded region ($M < 2.5 \text{ TeV}$) was not computed with the Monte Carlo; there the diffusion solve puts the lower edge of the excluded

band at 1–2 TeV. (b) The escape fraction. Data: figures/fig1_data_v2.json from supernova/sn_deep/transport_mc.json; script figures/make_fig1_v2.py.

Lower edge of the band. The diffusion solve gives $L_{\text{esc}}/3 \times 10^{52} = 1.2$ (cold, NN only) to 3.9 (hot) at $M = 1$ TeV, rising to $\times 4$ –30 at $M = 2$ –2.5 TeV; with degradation (Monte Carlo, $M \geq 2.5$) the cold NN -only case is only $\times 1.4$ –1.5 above the limit at 2.5–3 TeV. **The excluded band extends down to 1–2 TeV**, profile dependent. v1's criterion (" $\lambda_\phi \leq \lambda_\nu$, so ϕ carries a neutrino species' luminosity") was wrong: as for ν_μ, ν_τ , the ϕ energy sphere ($t_{\text{abs}} < t_{\text{diff}}$; 10.5–15 km, $T = 16$ –42 MeV at $M = 1$ –3 TeV) lies far inside the scattering sphere ($\tau = 2/3$ at 15–24 km, $T = 4$ –19 MeV), and a blackbody at the energy sphere overestimates L_{esc} by **18–32 $\times$** because the flux is diffusion-limited by the scattering atmosphere outside it (Raffelt 1996). Below ~ 1 TeV a hydrodynamic treatment is needed; that region lies outside the band of interest.

7. Bounds

Criteria: $\epsilon < 10^{19}$ erg g $^{-1}$ s $^{-1}$ at (30 MeV, 3×10^{14} g cm $^{-3}$, $Y_p = 0.3$) (Raffelt point) and $L_\phi(1\text{s}) < 3 \times 10^{52}$ erg/s (alternative 5.7×10^{52}) on two parametrized 1-s profiles from published figures ("cold": $1.41M_\odot$, $T_{\text{max}} = 31.5$ MeV; "hot": $1.33M_\odot$, 42.5 MeV). No public 1-s profile could be obtained (Appendix C); the $\times 1.35$ spread between profiles is the leading "profile" error, and both kinds of result are quoted with equal weight.

Final numbers (TeV):

Quantity	universal operator (1.1)	OP's operator (1.2)
central: NN ($K_T^{\text{FD}} = 7$ –10) + π ($Y_\pi = 1\%$), profiles, with transport	5.3 (cold) / 7.9 (hot) , i.e. $\gtrsim 6$; 4.3 / 6.7 for $L < 5.7 \times 10^{52}$	3.8 / 5.8 , i.e. ≈ 4.5 (0.72$\times$)
central, Raffelt point (free streaming)	8.0–8.2	5.6–5.8
NN only , profiles, with transport	3.5 / 6.0	2.8 / 4.9
NN only , Raffelt point	4.6–4.8	3.9–4.0
floor : traceless channel only, profiles / Raffelt point	2.2–3.9 / 2.9 ($\times 0.9$ –0.95 with transport)	same
lower edge of the excluded band	1–2 (profile dependent)	1–2
OP 2008 formula, same profiles / Raffelt point	13.2 / 16.9 / 14.8	same

The OP column rescales the universal one by the channel ratios of Sections 3 and 5 (NN rate $\times 0.35$ –0.45, pion rate $\times 0.22$; pions supply 63–75% of L in the central case), i.e. $M \times 0.71$ –0.74 (central) and $\times 0.8$ (NN only). **Standard statement:** for the universal (symmetron) coupling, $M \gtrsim 6$ TeV (5.3–7.9 across profiles), 3.5–6 TeV without pions, floor 2.2–3.9 TeV; for OP's literal operator $M \gtrsim 4.5$ TeV (3.8–5.8); the excluded region is a band from 1–2 TeV up to M_{min} . The number to defend in print is 5–6 TeV (universal). OP's estimate, applied to either operator, overstates the emissivity by 10–100 $\times$ (with pions / NN only).

Uncertainty budget for the central universal bound (factors on M_{min} , roughly independent, ranked):

rank	input	range	effect on M_{min}
1	Y_π - and in-medium pion physics (2024 mass shift, Δ -hole propagator, blocking 0.4–0.7)	$\times 0.3$ –3 in rate	$\times 0.75$ –1.3
2	PNS profile (cold vs hot; one 1-s snapshot; criterion 3 vs 5.7×10^{52})	$\times 1.35$ between profiles; $\times 0.85$ criterion	$\times 0.8$ –1.2
3	in-medium NN (m^* , T -matrix, LPM) on the trace channel	$\times 0.5$ –2 in rate	$\times 0.85$ –1.2 (NN only); $\times 0.95$ –1.05 (central)
4	K_T : finite- ω short-range model dependence	10–13 (MB)	$\times 0.95$ –1.02
5	transport (degradation, re-absorption; production spectra $\pm 30\%$)	Monte Carlo done	$\times 0.97$ –1.03
6	two-body pion bremsstrahlung, $\pi\pi \rightarrow \phi\phi$, Δ	+10–15% in rate	$\times 1.03$
—	operator identity (universal vs nucleon-only)	—	$\times 0.72$

8. Cosmological bound: thermal relic and ΔN_{eff}

The same operator keeps ϕ in thermal contact with the early-universe plasma, where the symmetron is in its symmetric phase with $m_{\text{eff}}^2 = (\rho - 3p)/M^2 \ll T^2$: one light real scalar. v2 used an instantaneous-decoupling estimate; v2.1 replaces it with a two-moment Boltzmann calculation for n_ϕ and ρ_ϕ (cosmology/neff_proper/ in the repository), integrated from 50 GeV to 5 MeV with the Saikawa-Shirai $g_{**}(T)$. The pair-production rate is $\gamma = (16\pi M^4)^{-1} \int d^4q (2\pi)^{-4} \Pi_\phi^2(q) \theta(q^2)$. Above the QCD crossover it is carried by the gluon trace anomaly $(b_0\alpha_s/8\pi)G^2$ and heavy quarks. Below it, $\pi\pi \rightarrow \phi\phi$ dominates, with the vertex $\langle 0|\Theta|\pi\pi\rangle = s + 2m_\pi^2$ of the low-energy theorem (Voloshin-Zakharov; Donoghue-Gasser-Leutwyler), Omnès-unitarized, plus K and η (the same vertex as the pion channel of Section 5). For $M = 5$ TeV, Γ/H stays at 0.2–0.6 from about 100 to 300 MeV. So ϕ never decouples sharply: it stays partly coupled through the crossover and shares in its entropy release, which the instantaneous criterion misses.

M (TeV)	3	4	5	6	8	10
ΔN_{eff} , central	0.35	0.30	0.24	0.17	0.094	0.067
band	0.30–0.38	0.21–0.35	0.14–0.32	0.10–0.27	0.07–0.16	0.056–0.10

Band: the envelope over hadronic-rate variants (LO ChPT to $\times 3$), QGP rate $\times 0.5$ –3, the α_s scale, T_c and crossover width, $g_{**} \pm 1\sigma$, and energy exchange with and without elastic scattering.

Limits. The combined BBN (D, ^4He) + Planck + ACT + SPT + DESI DR2 analysis gives $N_{\text{eff}} = 2.990 \pm 0.070$, i.e. $\Delta N_{\text{eff}} < 0.107$ at 95% (Goldstein & Hill 2026). Against it the calculation excludes $M \lesssim 7.5$ TeV (5.9–9.6 across the band). CMB-only limits give 6.8 TeV (< 0.13 , CMB-SPA) and 6.1 TeV (< 0.16 , ACT DR6), and the pre-2025 Planck+BAO limit (< 0.30) gives 4.1 TeV. BBN- and CMB-epoch values coincide, since ϕ decouples at $T \gg 1$ MeV.

Uncertainties. The dominant one is the $\pi\pi$ trace form factor at $\sqrt{s} \approx 0.5$ –1 GeV (± 1.7 TeV in M). A dispersive two-channel ($\pi\pi$, $K\bar{K}$) form factor would reduce it. QCD-crossover modelling moves M by only ± 0.3 TeV.

Caveats.

- *Reheating*. The bound assumes a standard thermal history. The thermal value of ΔN_{eff} is reached for $T_{\text{RH}} \gtrsim 200$ MeV at $M = 4$ –5 TeV, but only for $T_{\text{RH}} \gtrsim 1$ –3 GeV at $M = 6$ –8 TeV, where QGP-phase production matters. The excluded M is therefore 7.5 TeV for $T_{\text{RH}} \gtrsim 3$ GeV, 7.3 at 1 GeV, 6.2–6.6 at 0.2–0.5 GeV and 4.5 at 100 MeV, below the supernova bound. BBN itself needs only $T_{\text{RH}} \gtrsim 5$ MeV, and late entropy injection also weakens the bound.
- *Operator*. The bound applies to the universal coupling only. For (1.2) the pion channel is absent and $\Delta N_{\text{eff}} \approx 0.05$ –0.07, which is not excluded.
- *Relativistic relics*. The relics must be relativistic at recombination, which holds for a symmetron vacuum mass $\lesssim 10^{-3}$ eV.

Cosmology is thus the stronger bound under standard assumptions. The supernova bound of Section 7 is the one that holds for any thermal history.

9. Relation to the literature, and what is new

The 2007 unparticle papers. A $d_U = 2$ scalar unparticle coupled to $\bar{N}N$ is a $\phi\phi$ pair with the OP operator (Section 2), so the supernova bounds on $d_U = 2$ scalar unparticles are bounds on (1.2). Hannestad, Raffelt and Wong (arXiv:0708.1404) treat the vector coupling $\bar{N}\gamma_\mu N \mathcal{O}_U^\mu$, argue that for equal couplings to all nucleons bremsstrahlung is suppressed in the nonrelativistic limit "just as for $e^-e^- \rightarrow e^-e^-\gamma$ " so that quadrupole radiation with a factor v^4 dominates, and scale the graviton-bremsstrahlung rate of Hanhart et al. to $d = 1, 3/2, 2$. Dutta and Goyal (arXiv:0712.0145) treat the scalar coupling $(\lambda_S/\Lambda_U^{d_U-1})\bar{f}f\mathcal{O}_U$ explicitly, note that the leading $\pm 1/2m_N\omega$ terms from the outgoing and incoming legs "cancel in pairs", and compute the one-pion-exchange rate in degenerate matter for $d_U = 1$ –2.4, recovering Ishizuka-Yoshimura at $d_U = 1$. Freitas and Wyler (arXiv:0708.4339, v3) give the factorized soft-radiation rate for scalar and vector couplings with the same velocity scaling. **So the quadrupole-level correction to OP was available in 2007, for the OP operator.** Numerically, Dutta-Goyal's Fig. 1 gives $\lambda_S(d_U = 2, \Lambda_U = m_Z) \approx (1.5) \times 10^{-6}$ (read off the figure; their text quotes only $d_U = 1$ values), which with $\lambda_S/\Lambda_U = m_N/\sqrt{2}M^2$ is $M \approx 3.5$ –8 TeV, consistent with our NN -only bound for (1.2) (2.8–4.9 TeV; floor 2.2–3.9) within the reading uncertainty and the difference between their degenerate-OPE and our data-driven treatment.

Other prior art. The external-leg cancellation for mass-coupled scalars: Ishizuka-Yoshimura, Arndt-Fox, Keung-Ng-Tu-Yuan and Tu-Ng (Goldstone pairs), Dev et al., Balaji et al., Hardy et al., Joseph et al. (single CP-even scalars). The traceless channel is HPRS's theorem; the energy-sphere logic is Raffelt's (1996). **New here, as far as our search found (please tell us of earlier work):** (i) the exact pair identity (2.1)–(2.2) with the κ -weights 2/105 and 68/315 and no cross term; (ii) the time-delay form of the trace channel, its 0.1–0.5% check with three realistic potentials and its finite- ω thermal evaluation; (iii) the pion-induced pair channel; (iv) the transport treatment; (v) the explicit statement of which operator is bounded, with both bounds. We drop v1's headline "OP overstated by 2.5–5 $\times$ ": the headline is a calculation of $\phi\phi$ pair emission that combines conservation, the trace channel, pions and transport, with the operator stated; we have not found an earlier one. Appendix D condenses the literature search.

10. Conclusion, and what an expert should check first

The NN rate is the medium's stress spectral function: a traceless piece fixed by on-shell data and a larger trace piece fixed at $\omega \rightarrow 0$ by the phase-shift energy derivative and evaluated here at finite ω with realistic potentials. Pion absorption restores an unsuppressed channel, and transport in the semi-trapped regime costs 2–15% in M . The universal bound is $M \gtrsim 6$ TeV (5–8 across profiles; 3.5–6 without pions; floor 2.2–3.9), OP's literal operator gives ≈ 4.5 TeV (3.8–5.8), and the excluded band starts at 1–2 TeV. For a standard thermal history the ΔN_{eff} calculation of Section 8 excludes $M \lesssim 7.5$ TeV (5.9–9.6) for the universal coupling. Other stellar systems do not compete (Appendix B).

To check, in order: (1) the identity $\langle f|S|i \rangle = \langle f|2V + \tau V'|i \rangle$ and its limit $(1 + k d/dk)T \rightarrow e^{2i\delta} \delta'(k)$ (one page; trace_tables.py diagonal); (2) the κ -weights and the absence of a cross term in (2.2); (3) the $\pi^-p \rightarrow n\phi\phi$ matrix element in one basis (pion-line dominance, vertex sign $2m_\pi^2 + s$), the blocking factor, and Y_{π^-} with the 2024 mass shift; (4) whether the finite- ω enhancement survives with a soft $N^3\text{LO}$ potential fitted to data (our LO check says $\sim 0.8\times$); (5) the fits (4.1), (5.1) on a real 1-s profile with the luminosity criterion over 1–10 s. **Two concrete asks:** a 1-s $T(\tau), \rho(\tau), Y_e(\tau)$ (and Y_π) profile from the Garching Heinlein2023/hydroFiorilloEtAl23 set or an SFHo-18.8 model, and a look at the pion-channel matrix element.

Appendix A. Reconciliation of derivations A and B (condensed from v1 §3.3)

1. **Traceless rate.** A's "quadrupole" factor $[(\text{Tr}A)^2 + 2A:A]$ equals $2A^T:A^T + \frac{5}{3}(\text{Tr}A)^2$ with $\text{Tr}A = -m\omega \neq 0$; the extra piece is the $\kappa^4(\text{Tr}X/3)^2$ part of the trace channel. With the pure traceless weight A gives 7.0×10^{20} (MB) against B's 6.9×10^{20} at 30 MeV; the advisor's "1.6 $\times$ " was bookkeeping. The energy prescription (initial vs mean) matters at 9%; "final energy" is $\times 2.9$, driven by the 1S_0 resonance, and is disfavoured by HPRS's $O(\chi^2)$ argument.
2. **Trace channel.** A's thermal monopole used the weight $\int \kappa^2 d\kappa = 1/3$ instead of $(1 - \kappa^2/3)^2 \rightarrow 68/315$ and omitted the $-2\kappa^2/3$ interference. Corrected, A's finite- ω Reid68/Malfliet-Tjon 1S_0 evaluation gave $K_T = 8$ –14 on the pure-traceless baseline, against B's soft-limit 6.4; v1 adopted 6.5 (4–14) and flagged it as "probably low". Section 3 settles this: the exact AV18/Reid93/NijmII finite- ω evaluation gives 12 (10–13), i.e. A's range was right and B's soft limit was the low end. A's Reid68/MT numbers are superseded.
3. **Relativistic OPE (A, ope.py).** Full tree-level rate over A's Q-term: 0.65 in the soft region, 1.6 (nn) and 3.7 (np) over full phase space; emission from the pion line alone is 4–12 $\times$ the Q-term but interferes destructively with the legs. Consistent with $O(\omega^0)$ terms several times the soft pole.
4. **Pion channel.** A uses the s - and u -channel nucleon poles plus the pion line with vertex $(2k \cdot k' - 4m_\pi^2)/M^2$ (pion not rescaled); B adds a $\phi\phi\pi NN$ contact with the πNN vertex rescaled and vertex $-(2m_\pi^2 + s)/M^2$. Only the sum of diagrams is basis independent: in A's basis the pion line dominates and interferes destructively with the legs (30 MeV: pion line 7.1×10^3 , legs 1.0×10^3 , total 4.6×10^3 MeV $^2/M^4$); in B's all four diagrams add. The wrong-sign vertex $2m_\pi^2 - s$ would give $0.37\times$. There is no double counting with the NN channel (virtual pions). Per-pion rates agree to $\leq 5\%$ at $T = 20, 30, 40$ MeV before blocking; the " $\times 1.7$ " of the v1 reconciliation was entirely B's neutron blocking factor, now adopted as $B(T)$ in (5.1).
5. **Degeneracy.** B's traceless factor 0.85–0.88 was Monte Carlo noise; importance-sampled runs give 0.91–1.00 (A's 0.99 is right).

Appendix B. Other stellar bounds and the broken-phase caveat (condensed from v1 §7)

- **Neutron-star cooling.** Only the degenerate nn quadrupole contributes (the trace channel is $(E_F/\omega)^{-2}$ -suppressed); $\epsilon \propto T^6 M^{-4}$ against T^8 for modified Urca. Without the in-medium threshold, equality with modified Urca at $M = 0.5, 1.1, 1.5$ TeV for $T_0 = 1, 0.2, 0.1$; with the pair threshold $2m_{\text{eff}} \approx (80\text{--}90) \text{ keV} (\text{TeV}/M) \gtrsim T_{\text{core}}, L_\phi/(L_\nu + L_\gamma) \leq 0.1$ for $M \geq 1$ TeV before superfluid suppression. Not competitive; no shallow- T enhancement as in the linear case of Fiorillo et al. (2025).

- **Red giants, HB stars, Sun, white dwarfs.** Scalar charge α mass, so the ion-ion dipole vanishes; the quadrupole is $(T/m)^2$ -suppressed and the pair phase space $\propto T^6$; $M \gtrsim 0.1$ – 0.2 TeV at most.
- $\gamma\gamma \rightarrow \phi\phi$. For the conformal coupling $\xi_F \sim \alpha/\pi$, so OP's $M_*\xi_F^{-1/2} \gtrsim 3$ TeV becomes $M \gtrsim 0.3$ TeV.
- **Broken phase.** Where $\rho < \rho_c = \mu^2 M^2$, $\langle \phi \rangle = v(\rho) \neq 0$ and a linear Yukawa $g_N = m_N v / M^2 \lesssim 10^{-14}$ appears for the laboratory band $v/M \approx 10^{-14}$ – 10^{-11} . Every stellar interior is in the symmetric phase, so linear-scalar SN and NS bounds do not apply. Solar symmetron production by photon conversion (Banks, Davis, Visinelli 2026) requires ρ_c above the tachocline density, far above our band. For the opposite coupling sign (Balkin et al. 2025) dense matter sources ϕ and the induced linear coupling in the core gives $M_\phi \gtrsim 10^7$ TeV.

Appendix C. Numerical details

- **Inputs.** Both codes evaluate $Q_{NN} = \sum_{ab} \frac{n_a n_b}{1+\delta_{ab}} \langle v \int d\omega \omega G(\omega)(\mathbf{p}'/\mathbf{p})(\text{Pauli}) \rangle$ with spin-averaged cross sections. A: PWA93 phases from nn-online, Fermi-Dirac by Monte Carlo; B: approximate PWA93/SAID tables to 800 MeV, Maxwell-Boltzmann plus a degeneracy correction; σ and $\sigma^{(2)}$ agree to $\leq 10\%$ below 350 MeV. Non-degenerate traceless closed form: $Q_T \simeq \frac{4}{11025\pi^4 M^4} \sum_{ab} \frac{n_a n_b}{1+\delta_{ab}} \langle v \sigma^{(2)}(E) E^5 \rangle$, $Q_T/Q_{OP} \approx (1.4$ – $1.7)(T/m_N)^2$. nn from pp with Coulomb removed (charge-independence and charge-symmetry breaking ignored); A's PWA93 cross sections held constant above 350 MeV; B's tables within 10% of PWA93 in σ ; relativistic corrections 10–50% at $p \gtrsim 250$ MeV. The potential-model solver reproduces PWA93 np phases to $\lesssim 0.5^\circ$ for $T_{lab} \leq 300$ MeV (1S_0 , 3S_1 , 3D_1 , ϵ_1 , all P, D, F, G waves). A sign error in the asymptotic matching of the solver (Cramer's rule for the cosine amplitude) flipped all phases in an early run and was fixed before any number in this note was produced.
- **Trace tables.** Grid $E_{cm} = 0.5$ – 460 MeV, all channels to $J = 4$; $\sigma_{tr}(E_f, E_i) = w\pi H(E_f, E_i)/(k_i k_f)^2$; soft limit $w\pi \sum (2J+1) \frac{1}{4} \text{Tr}|dS/dk|^2$ from the same potential. Thermal integrals: $\int d^3k P(k) \int d\omega \omega^4 v' (68/315) \sigma_{tr}$ (Maxwell-Boltzmann), degeneracy by importance-sampled Monte Carlo ($N = 4 \times 10^4$).

Table C1. Theorem check: exact $\langle f|2V + \mathbf{r}V'|i \rangle$ on the diagonal $E_f = E_i$ over the Wigner-time-delay form (3.2), and $c_{tr}^2 = \sigma_{tr}/\sigma_{el}$, for AV18 (Reid93 and NijmII agree to the digits shown in the ratio; their c_{tr}^2 at 148 MeV are 2.90/2.60 and 2.81/2.68).

E_{cm} (MeV)	T_{lab}	ratio, nn / np	c_{tr}^2 , nn / np
9.5	19	1.005 / 1.003	0.41 / 0.70
24	48	1.005 / 1.005	1.36 / 1.31
48	96	1.002 / 1.002	2.57 / 2.28
76	152	1.001 / 1.001	3.02 / 2.70
100	200	1.0007 / 1.0007	3.07 / 2.77
148	296	1.0005 / 1.0005	3.00 / 2.75
176	352	1.0005 / 1.0005	3.04 / 2.78

Table C2. Effective trace/traceless ratio with Fermi-Dirac blocking (AV18 finite- ω), from `nn_emissivity_new.json`.

$(T [\text{MeV}], \rho [\text{g cm}^{-3}], Y_p)$	(20, 3×10^{14} , 0.3)	(30, 3×10^{14} , 0.3)	(40, 3×10^{14} , 0.3)	(31, 2.8×10^{14} , 0.12) cold peak	(42, 2.6×10^{14} , 0.13) hot peak	(30, 10^{14} , 0.3)	(25, 4.4×10^{14} , 0.22)
K_T^{MB}	11.6	12.2	12.1	11.4	11.3	12.2	11.9
K_T^{FD}	6.3	8.1	9.1	7.0	8.4	10.5	5.6
ϵ_{NN}^{FD} (1 TeV)	4.9×10^{20}	5.4×10^{21}	2.7×10^{22}	4.8×10^{21}	2.7×10^{22}	2.4×10^{21}	2.0×10^{21}

- **Chiral LO check.** Local LO potential, $\delta_{R_0} = \exp[-(r/R_0)^4]/(\pi\Gamma(3/4)R_0^3)$, $R_0 = 1.0$, 1.2 fm; $C_{1S_0} = -355/-430$, $C_{3S_1} = -102/-349$ MeV fm³ fitted at 10 MeV; LO 1S_0 at 300 MeV is $+30^\circ$ against -4.5° (PWA93), so only ratios are used.
- **Pion rate.** Tree level, both bases; $Y_\pi(T) = Y_0 \exp[-(m_\pi - 80)(1/T - 1/37)]$ capped at $5Y_0$; $\epsilon_\pi = 0$ for $\rho < 10^{12}$.
- **Transport.** $N = 2000$ pairs per point, sub-steps ≤ 1 km with direction kept between sub-steps (an earlier version added an artificial 3-km scattering atmosphere; fixed). Production spectra: pairs $\sim \Gamma(6, T) (NN)$, $\Gamma(12, 190/12)$ (pion), split flat between the two ϕ . Diffusion: flux-limited $D = c\lambda/(3 + \lambda|u|/u)$, Marshak boundary at 25 km.
- **Profiles.** Parametrized "cold" and "hot" from published figures (supernova/sn_A/profile_bound.py). What was tried for public profiles (26 Sep 2026): the Garching CCSN archive lists Bollig2016_radial_profiles, Heinlein2023/hydroFiorilloEtAl23 (1D PNS cooling, SFHo/LS220, to 10 s), Ringler2020, Lella2026 and others, all HTTP 401 (request-based access, contact H.-T. Janka); the arXiv pages of Caputo-Raffelt-Vitagliano, Lella et al., Carenza et al., Fiorillo et al. 2023, Fischer et al. 2016, Hardy et al., Joseph et al. carry no ancillary profiles; the Nakazato database and Zenodo PNS-cooling records hold neutrino spectra only.
- **Figure 2 of v1** (fig2_Mmin_vs_Ypi.png, free streaming, $K_T = 6.5$) remains on disk as an illustration of the $Y_\pi^{1/4}$ dependence; it is not updated here.

Appendix D. Literature search (2007 - Sept 2026), condensed

Paper	What it does	Supersedes OP for this operator?
Olive & Pospelov 2008	Weizsäcker-Williams estimate, 15 TeV, operator (1.2)	—
Hannestad, Raffelt & Wong 2007	vector unparticle; quadrupole (v^4) scaling; $d = 1, 3/2, 2$	quadrupole-level correction, vector coupling
Dutta & Goyal 2008	scalar $\tilde{f}fO_V$; leg terms cancel in pairs; OPE, degenerate; $d_V = 1$ – 2.4	quadrupole-level, operator (1.2), OPE
Freitas & Wyler 2007	scalar and vector unparticles, factorized soft radiation	quadrupole-level, operator (1.2)
Stadnik & Flambaum 2015; Hees et al. 2018; Banerjee et al. 2023; Bouley et al. 2023; Balkin et al. 2025; Baryakhtar et al. 2025; Bartnick et al. 2026; Banks et al. 2026	quote OP or do not treat SN energy loss	no

Paper	What it does	Supersedes OP for this operator?
Fiorillo et al. 2025; Hardy et al. 2025; Joseph et al. 2026; Dev et al. 2020; Balaji et al. 2022	linear CP-even scalars	no (cancellation noted)
Keung et al. 2014; Tu & Ng 2017	Goldstone pairs via Higgs portal	no; different operator, cancellation noted
HPRS 2001; Arndt & Fox 2002; Ishizuka & Yoshimura 1990	KK dilaton; saxion; dilaton	no; the traceless channel is HPRS's
Brax & Burrage 2014	disformal pairs, $M \gtrsim 92$ GeV	no; different operator

INSPIRE lists 362 papers citing OP; all titles were scanned and the relevant ones read. None includes the trace channel, the pion channel or transport for this operator.

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Files

Paths are relative to the repository root.

File	Contents
docs/sn_technical_note_v2.md	this note
figures/fig1_Lesc_vs_M.png, figures/fig1_data_v2.json, figures/make_fig1_v2.py	Figure 1, its data and script
supernova/sn_deep/pw_solver.py, trace_tables.py, trace_*.npz	coupled-channel solver, finite- ω trace tables (AV18, Reid93, NijmII; ω only for operator (1.2))
supernova/sn_deep/trace_thermal.py, trace_degeneracy.py, nn_emissivity_new.py/.json	thermal K_T , Pauli blocking, NN emissivity grid
supernova/sn_deep/chiral_lo_check.py/.out	LO chiral cross-check
supernova/sn_deep/transport_mc.py/.json, diffusion_1d.py	transport
supernova/sn_deep/twobody_estimate.py	two-body pion bremsstrahlung estimate
supernova/sn_deep/SUMMARY.md	running summary of the v2 calculations
cosmology/neff_proper/	ΔN_{eff} Boltzmann calculation (Section 8) and its summary
supernova/sn_A/, supernova/sn_B/	v1 source scripts